

Executive Summary

Following an architectural evaluation of the ParaLearn platform (NestJS backend & Next.js frontend), we recommend extracting the **Computer-Based Testing (CBT) engine as an autonomous microservice (cbt.pln.ng)**. This move resolves critical high-concurrency exam spikes that currently threaten core school operations, while unlocking a high-growth self-serve market for tutorial centres (JAMB/WAEC prep), independent tutors, and testing institutes without requiring full School Information System (SIS) onboarding.

1. Strategic & Operational Rationale

- **Resource & Traffic Asymmetry:** During exam weeks, thousands of students simultaneously hit servers with start requests, answer updates, 20-second proctoring heartbeats, and auto-submits. Decoupling isolates connection pools and ensures report card generation, bursary billing, and admin tasks run unaffected.
- **Product-Led Growth (PLG) & TAM Expansion:** Full school onboarding requires academic sessions, terms, classes, subject teachers, and grading scales—a major barrier for standalone examiners. Decoupling allows tutors to register in 30 seconds, create a test, and distribute PINs.
- **Dual-Access Synergy:** Existing ParaLearn schools retain Single Sign-On (SSO) and automatic score-sheet sync, while independent examiners get a lean, self-serve testing hall.

2. Platform Feature Audit & Extraction Roadmap

Overview of the current monolith domain boundaries and their target deployment profiles:

Domain / Module	Current Monolith Path	Target Architecture	Priority	Primary Driver
CBT & Proctoring	src/assessments, src/proctoring	Autonomous Microservice	P0 - IMMEDIATE	Extreme burst concurrency, independent B2B/B2C SaaS market.
Result Processing & PDFs	src/reports, src/Queue, src/pdf	Dedicated Worker Service	P0 - IMMEDIATE	CPU/RAM-heavy headless PDF booklet rendering (BullMQ + Puppeteer).
AI Lesson Planner (SabiNote)	src/llm, src/credits, sabinote	Standalone EdTech SaaS	P1 - HIGH	Already semi-decoupled; third-party LLM rate-limiting and credit billing.
Finance & Bursary	src/fees, src/invoices	Finance Microservice	P2 - MEDIUM	Financial ledger immutability, payment gateway webhooks (Paystack).
Core SIS (Classes, Roster, Terms)	src/classes, src/users, src/academic	Core Monolith	KEEP IN CORE	Central source of truth for institution structure and identity records.

3. Core Architectural Concept: The "Workspace" Abstraction

To cleanly support both ParaLearn institutional schools and independent tutors without maintaining duplicate codebases, the CBT service introduces an abstract **Workspace** model:

```
cbt.pln.ng | _____ ▼ ▼ [Mode A: ParaLearn School] [Mode B:
Standalone Exam Hall] • Auth: SSO via ParaLearn JWT • Auth: Self-Serve (Email / Google) • Workspace:
School Tenant • Workspace: Tutorial Centre / Tutor • Roster: Synced from School Classes • Roster: Ad-
hoc PINs / Public Link • Egress: Auto-feeds Term Report Cards • Egress: Instant Results / CSV Export
```

Mode A: ParaLearn School User (SSO)

School admin or teacher logs in via ParaLearn credentials. The CBT backend verifies the JWT against the shared signing authority and maps the session to an INSTITUTION workspace with existing classes and subjects preloaded.

Mode B: Standalone Exam Hall (Self-Serve)

Independent tutors sign up directly with Email or Google OAuth. Creates a STANDALONE_HALL workspace. Zero SIS setup required—tutors immediately create question banks and launch tests.

4. Frictionless Candidate Delivery: Roster vs. Exam PIN

Tutorial centres and walk-in lesson centers cannot manage permanent student accounts. The CBT microservice supports two distinct delivery strategies:

- **Roster Model (Schools):** Permanent student account (student ID + password). Results attach to academic transcripts and end-of-term report cards.
- **Exam PIN & Direct Link Model (Tutors & Tutorial Centres):** Tutors share a direct link (e.g. `cbt.pln.ng/take/JAMB-MOCK-26`). Candidates enter their **Full Name + One-Time Access PIN** and take the test immediately without account friction.

5. Decoupled Data Model (Prisma Schema Preview)

```
// — Core Tenancy ————— model CbtWorkspace { id
String @id @default(cuid()) name String // "Greenfield College" or "Success Point Tutors" type
WorkspaceType // INSTITUTION | STANDALONE_HALL externalId String? // Links to ParaLearn School.id (if
INSTITUTION) ownerId String exams Exam[] } enum WorkspaceType { INSTITUTION STANDALONE_HALL } // —
Exam & Anti-Cheating Configuration ————— model Exam { id String @id
@default(cuid()) workspaceId String title String durationMins Int accessType AccessType // ROSTER_ONLY
| PUBLIC_LINK | ACCESS_CODE accessCode String? // e.g. "PHY-MOCK-01" maxTabViolations Int @default(3)
shuffleQuestions Boolean @default(true) shuffleChoices Boolean @default(true) showResultAfter Boolean
@default(true) questions Question[] attempts ExamAttempt[] } // — Concurrency-Optimized Attempts
————— model ExamAttempt { id String @id @default(cuid()) examId
String candidateName String candidateId String? // Nullable for ad-hoc link takers startedAt DateTime
@default(now()) submittedAt DateTime? score Float? status String // IN_PROGRESS | SUBMITTED |
DISQUALIFIED antiCheatLogs Json? // Tab switches, window blur events, device info answers
AttemptAnswer[] }
```

6. High-Throughput Infrastructure & Data Sync Contract

- **Redis In-Memory State:**
 - **Active Timers & Locks:** Prevents multi-device logins and synchronizes countdowns accurate to milliseconds.
 - **Ephemeral Answer Buffer:** Caches in-progress candidate keystrokes and answers in Redis, eliminating thousands of database writes per minute. Answers are bulk-flushed to PostgreSQL only on final submission.
- **Inter-Service Communication:**
 - **Ingress:** Read-only gRPC or internal REST endpoints to pull class rosters from ParaLearn Core for school exams.
 - **Egress (Score Sync):** When an institutional exam concludes, the CBT service emits an `exam.completed` event to BullMQ/Redis. ParaLearn Core consumes this and executes an atomic `POST /scores/batch-save` to update report cards automatically.

7. Phased Implementation Roadmap

Phase	Timeline	Deliverables
Phase 1: Foundation & Schema	Weeks 1–2	Spin up isolated service repository and PostgreSQL instance with CbtWorkspace schema. Build question authoring (MCQ, True/False, Theory) and bulk Excel import.
Phase 2: Dual-Auth & Link Runner	Weeks 3–4	Implement ParaLearn JWT SSO + self-serve Auth (Google/Email). Build candidate exam runner at <code>cbt.pln.ng/take/:code</code> with tab-detection anti-cheat and localStorage crash recovery.
Phase 3: Core Sync & Migration	Weeks 5–6	Implement asynchronous score-sync back to ParaLearn Core. Migrate existing frontend <code>/RMS/cbt</code> and <code>/student/exam</code> routes to target the new microservice gateway.
Phase 4: Monetization & PLG Launch	Week 7+	Introduce Paystack credit wallet (pay-per-candidate top-up) and freemium tier (first 30 candidates free) for standalone tutors.

8. Action Items for CTO Sign-Off

1. **Boundary Alignment:** Confirm prioritization of CBT as Service #1, followed by Result Processing / PDF workers as Service #2.
2. **Repository Strategy:** Decide between Monorepo (Nx / Turborepo housing Core and CBT) vs. Polyrepo (dedicated `paralearn-cbt` repository).
3. **Event Bus Choice:** Approve using our existing Redis/BullMQ infrastructure for inter-service communication or upgrading to RabbitMQ.